

rotation increases. But if you want to increase the speed, you just cannot keep on increasing the voltage across its terminals, as if you go beyond the rated voltage, the internal coil will burn and the motor will be damaged. So, it is always better to operate around the rated voltage.

Current draw

Current draw is the amount of current that the motor draws from the power supply. For small DC motors, it is measured in in milliamps. The current drawn by a motor when no load is applied to it can be quite low. But when the load is applied, the current draw becomes three or 4 times, depending upon the load applied. Most of the DC motors you will encounter in the market are permanent magnet motors, and in all of them current drawn increases with the load applied. If you keep on increasing the applied load, a stage is reached when the motor shaft doesn't rotate at all. No more current flows through it. This condition is called motor stalling. Some motors are rated by the manufacturer by the amount of current they would draw when stalled.

Speed

The rotational speed of a motor is given in revolutions per minute (rpm). You can get DC motors having a normal operating speed of 10 rpm to those having operating speed of 7000 rpm. For robotic applications, you would need very low rpm motors, and it's unlikely that you would ever go beyond a 500 rpm motor in robotics.

Torque

Torque is the force applied by a revolving axle. The higher the torque, the larger is the force applied. You can actually test the torque of a motor by trying to rotate its axle when it is not powered by batteries. The more power you require, the more torque that particular motor has. If the torque of the motor is less compared to the load being applied, the motor would simply stall and consume a lot of current and dissipating a lot of heat.

3.3.6 Battery

If you want your robot be to be completely self-operated, and you do not want any wire coming out of the robot even for getting power, you need to provide on board batteries. Many different types of batteries are available in the market, NiCad or NiMH batteries being most common.

You can obtain higher voltages and current by connecting several cells